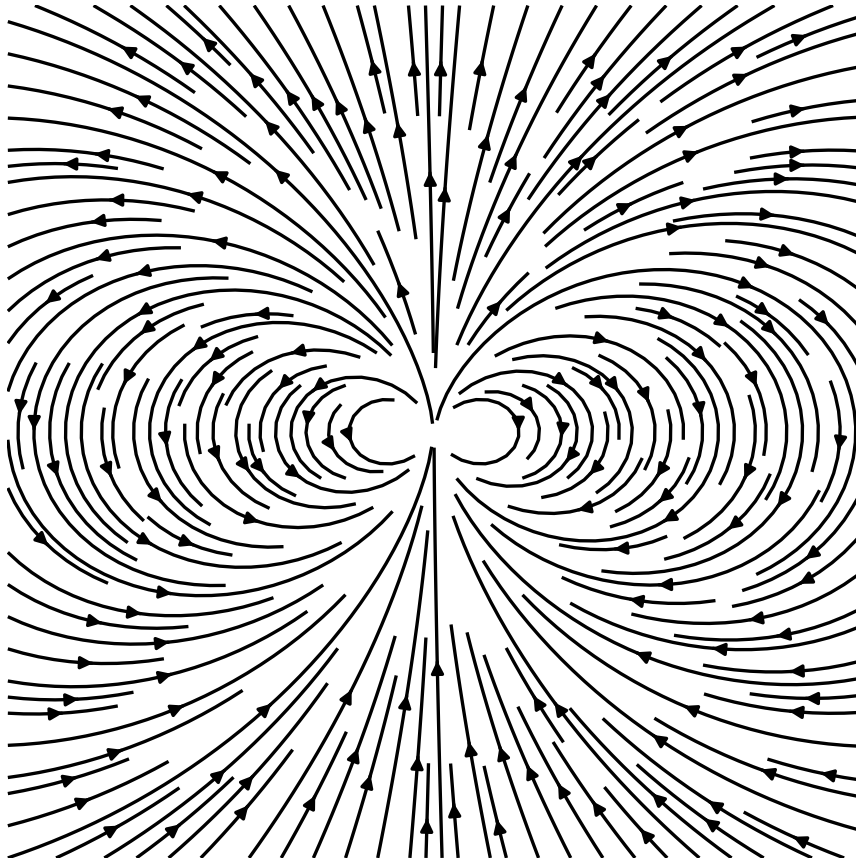




Lecture Notes on Electromagnetic Theory

PHYSICS 131

VERCIL JUAN
IV - BS Physics

1st Semester, AY 2026-2027

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} \\
\nabla \cdot \mathbf{B} &= 0 \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
\nabla \times \mathbf{B} &= \mu_0 \mathbf{j} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}
\end{aligned}$$

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Physics 131 - Electromagnetic Theory I

Course Information

Course Description

Description: Electrostatics in a vacuum, electrostatics in dielectric media, boundary value methods in electrostatics, electric currents, conducting media, magnetostatics in a vacuum, macroscopic and microscopic magnetism, Faraday's law of electromagnetic induction.

Credits: 3.0

Instructor Details

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Lecture Hours: WF, 8:30 - 10:00 AM

Course Requirements

You must submit two (2) notebook files, one (1) problem set, and take one sit-in midterm exam to complete the course. Notebooks and problem sets may be handwritten or typeset, but the submission must be a PDF file to be submitted through an online form. The submission link will be provided on the course webpage. You must submit your work by the deadline, otherwise the work will be considered a non-submission. The submission bin will close 48 hours after the deadline as a grace period, and late submissions within this period must be accompanied by a University-recognized excuse.

On-site attendance will be checked to comply with University attendance requirements.

Progress will be monitored by performing two types of checkpoint activities:

1. Notebook Entries
2. Problem Sets

Main References

The Syllabus for the course can be found here:

■ [COURSE SYLLABUS](#)

The main reference for this course is:

■ David J. Griffiths, [INTRODUCTION TO ELECTRODYNAMICS](#) 4th Edition (2013)

Reminders

Exam Dates

☐ Midterm Exam :

Problem Set Dates

☐

Notebook Submission Dates

☐

☐

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Unit 1

Electostatics

1.1 Review of Vector Calculus

This is basically just a summary of the **Physics 116 curriculum**, and as such, **I will be pasting an abridged copy of my notes there to here.** Anyway.

1.1.1 Vectors and Scalars

A **scalar** $a \in \mathbb{R}$ is a quantity with magnitude. For the purposes of this notes, scalars are denoted in italic script. A scalar is in the set of real numbers

$$a \in \mathbb{R} \quad (1.1.1)$$

An example of a scalar is the temperature in a room, T_0 , time t , and other single valued quantities.

A **vector** is a collection of scalars, usually indicating magnitude and direction. We represent vectors in math bold face. A vector of n dimensions is represented by n numbers, v_i , where i represents the i -th element of the vector. A vector can be a column vector or a row vector.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad \mathbf{v}^\top = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \quad (1.1.2)$$

An example of a vector is a particle's position in space

$$\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (1.1.3)$$

A tensor is a generalization of vectors. It is essentially an n dimensional (More commonly called rank) vector. This is generally beyond the scope of this notes, but I included it for generality. An example of a tensor is the stress energy tensor.

$$T = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \quad (1.1.4)$$

It's essentially a matrix, but more general. A matrix is a rank 2 tensor. A vector is a rank 1 tensor, while a scalar is a rank 0 tensor.

1.1.1.1 Vector operations

There are several operations that can be done with vectors and scalars. Here are some of them.

Vector addition For a vector $\mathbf{u}$ and $\mathbf{v}$, adding them is done by adding element-wise. The same is true for subtraction.

$$\mathbf{u} \pm \mathbf{v} = (u_i \pm v_i)\hat{\mathbf{e}}_i \quad (1.1.5)$$

Scalar multiplication A vector multiplied by a scalar is equivalent to multiplying all its elements with the scalar. As a shorthand, we can represent vectors and all its elements with one expression $\mathbf{v} = v_i$, where the index i runs over from 1 to n . Given $a \in \mathbb{R}$, a scalar multiplication is

$$a\mathbf{v} = \begin{bmatrix} av_1 \\ av_2 \\ av_3 \end{bmatrix} = av_i\hat{\mathbf{e}}_i \quad (1.1.6)$$

Dot Product The dot product of two vectors is the sum of the product of the i -th element of each vector. For example, given a vector $\mathbf{v} = v_i$ and $\mathbf{u} = u_i$, their dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = \sum_i u_iv_i \quad (1.1.7)$$

We will encounter a lot of these sums in general, so we introduce the *Einstein summation convention* where, if we see repeated indices, this implies that the expression is being summed over that index. As such, we can more conveniently define a dot product as

$$\mathbf{u} \cdot \mathbf{v} = u_iv_i \quad (1.1.8)$$

Cross Product The cross product is pretty hard to define, but in general, it is shown the vector which is the determinant of the matrix of the vector elements. But first, we introduce the following vectors

$$\hat{\mathbf{e}}_1 = \hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_2 = \hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_3 = \hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (1.1.9)$$

There are known as **elementary orthogonal basis vectors** of the vector space. Basically unit vectors of a single dimension. We can represent any vector in $\mathbb{R}^3$ in this notation.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_1\hat{\mathbf{i}} + v_2\hat{\mathbf{j}} + v_3\hat{\mathbf{k}} = v_i\hat{\mathbf{e}}_i \quad (1.1.10)$$

It must be noted that $\mathbf{v} = v_i(x, y, z)\hat{\mathbf{e}}_i$. In that definition, then, we often see the dot product defined as

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2)\hat{\mathbf{i}} - (u_1v_3 - u_3v_1)\hat{\mathbf{j}} + (u_1v_2 - u_2v_1)\hat{\mathbf{k}} \quad (1.1.11)$$

The Kronecker Delta and LeviCivita operators The Kronecker delta is an operator which basically turns on or of a product depending on it's indices. Formally, it is defined as

$$\delta_{ij} = \hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (1.1.12)$$

and the Levi-Civita symbol is defined to be 0 if two indices are the same, 1 if the indices are an even permutation of i, j, k and -1 if it is not.

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } ijk \text{ is } 123, 312, 231 \\ -1 & \text{if } ijk \text{ is } 132, 321, 213 \\ 0 & \text{if } i = j \text{ or } j = k \text{ or } i = k \end{cases} \quad (1.1.13)$$

with this, we can more succintly define the dot and cross product as

$$\mathbf{u} \cdot \mathbf{v} = \delta_{ij} u_i v_j \quad (1.1.14)$$

$$\mathbf{u} \times \mathbf{v} = \epsilon_{ijk} u_i v_j \hat{\mathbf{e}}_k \quad (1.1.15)$$

Einstein Summation Notation To prevent clunkiness of equations, we oftem omit the summation symbol. By convention, when an index appears **twice** in a single term (once as a superscript and once as a subscript), it is **implicitly summed over** all values of that index.

$$\sum_i a_i b^i \equiv a_i b^i \quad (1.1.16)$$

The summation symbol $\sum$ is ommited. This applies to the tensor notation listend above. Sometimes, it is also not done with superscripts, but the summation convention applies nonetheless.

$$\sum_i a_i b_i \equiv a_i b_i \quad (1.1.17)$$

1.1.2 Vector Properties

1.1.2.1 The Vector Norm

The vector norm is the *length* of the vector, so to speak. It is defined as

$$\|\mathbf{v}\| = \sqrt{\mathbf{v}^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}} \quad (1.1.18)$$

Note that we define the square of a vector to be its dot product with itself. $\mathbf{v}^2 = \mathbf{v} \cdot \mathbf{v}$

1.1.2.2 Vector Axioms

Technically, a vector is really just defined as an element of a vector space, which is an abstract space which obeys the following properties¹.

¹This comes from the Math 40 curriculum

Definition 1.1 : Vector Axioms

Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{V}$. The space $\mathbb{V}$ is a vector space if the following conditions hold:

These operations are defined as follows:

1. Vector addition, $+: V \times \mathbb{V} \rightarrow \mathbb{V}$
2. Scalar multiplication, $\cdot: \mathbb{R} \times \mathbb{V} \rightarrow \mathbb{V}$

and these conditions must hold

1. Closure under addition, $\mathbf{u} + \mathbf{v} \in \mathbb{V}$
2. Commutativity, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. Associativity, $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{v} + (\mathbf{u} + \mathbf{w})$
4. Additive identity (zero vector $\hat{0}$), $\forall \mathbf{v} \in \mathbb{V}, \exists \hat{0} \in \mathbb{V} : \mathbf{v} + \hat{0} = \mathbf{v}$
5. Additive inverse, $\forall \mathbf{v} \in \mathbb{V}, \exists -\mathbf{v} \in \mathbb{V} : \mathbf{v} + (-\mathbf{v}) = \hat{0}$
6. Closure under scalar multiplication, $\forall a \in \mathbb{R} \wedge \mathbf{v} \in \mathbb{V}, a\mathbf{v} \in \mathbb{V}$
7. Distributivity of scalar multiplication under vector addition, $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
8. Distributivity of scalar multiplication under scalar addition, $\forall a, b \in \mathbb{R}, (a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$
9. Associativity of scalar multiplication, $a(b\mathbf{v}) = (ab)\mathbf{v}$
10. Existence of multiplicative identity, $\forall \mathbf{v} \in \mathbb{V}, \exists 1 : 1 \cdot \mathbf{v} = \mathbf{v}$

1.1.2.3 Vector derivatives and Integrals

To take the derivative of a vector, one must apply the derivative on all of its components, just like normal derivatives. Chain rule and other calculus methods still apply.

$$\frac{d\mathbf{v}}{dt} = \begin{bmatrix} \frac{dv_1}{dt} \\ \frac{dv_2}{dt} \\ \frac{dv_3}{dt} \end{bmatrix} = \frac{dv_i}{dt} \quad (1.1.19)$$

Similarly, the same applies for integration, along with path integrals and the like which will not be part of this section.

$$\int \mathbf{v} dt = \begin{bmatrix} \int v_1 dt \\ \int v_2 dt \\ \int v_3 dt \end{bmatrix} = \int v_i dt \quad (1.1.20)$$

1.1.3 Vector Transformations

1.1.3.1 Linear transformations

Consider a vector

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_i \hat{\mathbf{e}}_i$$

A linear transformation is the operation

$$\mathbf{v}' = A\mathbf{v} \quad (1.1.21)$$

where A is an $n \times n$ matrix. Linearity in this case means

$$A(a\mathbf{u} + b\mathbf{v}) = aA\mathbf{u} + bA\mathbf{v} \quad (1.1.22)$$

The matrix A is described as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad (1.1.23)$$

Notation A matrix is denoted in index notation as

$$A = a_{ij}$$

It has a few defining properties, but we consider the determinant and the trace. The trace is defined as follows

$$A^\top = a_{ji}$$

It is assumed that the reader already knows the determinant.

Multiplying matrices A and B are defined as follows

$$AB = a_{il}b_{lj}$$

A general 2×2 linear transformation is described in matrix form as

$$\mathbf{v}' = A\mathbf{v} \quad (1.1.24)$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad (1.1.25)$$

As such,

$$x' = ax + by \quad y' = cx + dy \quad (1.1.26)$$

1.1.3.2 Eigenvectors, Eigenvalues

An **eigenvector** is a vector whose direction is unchanged by a transformation.

Definition 1.2 : Eigenvector

$$A\mathbf{v} = \lambda\mathbf{v} \quad (1.1.27)$$

Where λ is known as the **eigenvalue**. To find an eigenvalue, consider

$$(A - \lambda\mathbb{1})\mathbf{v} = 0 \quad (1.1.28)$$

where $\mathbb{1}$ is an identity matrix

$$\mathbb{1} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \quad (1.1.29)$$

For a non trivial vector, we have

Definition 1.3 : Characteristic equation

$$\det(A - \lambda\mathbb{1}) = 0 \quad (1.1.30)$$

This is the **characteristic equation**. Once λ is known, (1.1.30) is solvable.

1.1.3.3 Translation

Translation is different from ordinary matrix multiplications, because

$$\mathbf{r}' = \mathbf{r} + \mathbf{a} \quad (1.1.31)$$

where $\mathbf{a}$ is another vector. Doing this moves or *translates* the vector.

For example

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix} \quad (1.1.32)$$

This is not linear, because generally

$$T(a\mathbf{v}) \neq aT(\mathbf{v}) \quad (1.1.33)$$

It is instead an *affine transformation*

A general affine transformation is

$$\mathbf{r}' = A\mathbf{r} + \mathbf{b} \quad (1.1.34)$$

1.1.3.4 Rotation

2 Dimensions A rotation in two dimensions through an angle θ is represented by

Definition 1.4 : Rotation Matrix ($SO(2)$)

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad (1.1.35)$$

and, the transformation is defined by

$$\mathbf{v}' = R(\theta)\mathbf{v} \quad (1.1.36)$$

thus

$$x' = x \cos \theta - y \sin \theta \quad (1.1.37)$$

$$y' = x \sin \theta + y \cos \theta \quad (1.1.38)$$

The defining property of a rotation matrix is

$$R^\top R = \mathbb{1} \quad (1.1.39)$$

and

$$\det(R) = 1 \quad (1.1.40)$$

The set of all such 2×2 matrices form the special orthogonal group in 2 dimensions $SO(2)$.

Rotations preserve lengths and angles during their transformations.

$$\|\mathbf{v}'\| = \|\mathbf{v}\|$$

3 Dimensions In three dimensions, rotations are represented as much the same.

$$\mathbf{v}' = R\mathbf{v}$$

with the same properties as $SO(2)$. All such matrices form the group known as $SO(3)$. Unlike $SO(2)$, a 3D rotation generally has a rotation axis

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1.1.41)$$

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \quad (1.1.42)$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \quad (1.1.43)$$

1.1.4 Field

A *field* is a function that assigns a value to every point in a defined space. In other words, a field takes the position of a point in space as its input and returns some physical or mathematical quantity associated with that point.

For a three-dimensional space, we can specify the position of a point using the position vector

$$\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

The field is then a function of this position vector. Depending on the type of quantity being assigned, we distinguish between scalar fields and vector fields.

1.1.4.1 Scalar field

A scalar field assigns a single scalar value to every point in space. For example, the temperature distribution in a room can be represented by

$$T(\mathbf{r}) = T(x, y, z).$$

Here, the input $\mathbf{r}$ specifies a particular point in space, while the output $T(\mathbf{r})$ is the temperature at that point.

Thus, a scalar field can be thought of as a rule that associates one number with every point:

$$\mathbf{r} \mapsto T(\mathbf{r}).$$

For example,

$$T(1, 2, 3) = 300 \text{ K}$$

would mean that the temperature at the point $(1, 2, 3)$ is 300 K.

We often write a scalar field as

$$\phi(\mathbf{r}),$$

where ϕ denotes the field and $\mathbf{r}$ specifies the position at which the field is evaluated.

1.1.4.2 Vector field

A vector field assigns a vector to every point in space. For example, the velocity of a fluid flowing through a region of space can be represented by

$$\mathbf{v}(\mathbf{r}) = \mathbf{v}(x, y, z).$$

Unlike a scalar field, which returns a single number, a vector field returns a vector whose components are themselves functions of position.

We can therefore write

$$\mathbf{F}(\mathbf{r}) = \begin{bmatrix} F_1(\mathbf{r}) \\ F_2(\mathbf{r}) \\ F_3(\mathbf{r}) \end{bmatrix} = F_i(\mathbf{r})\hat{\mathbf{e}}_i$$

where each component

$$F_i(\mathbf{r})$$

specifies the corresponding component of the vector at the point $\mathbf{r}$.

Thus, at each point in space, the field provides both a direction and a magnitude.

The important distinction is therefore that $\mathbf{r}$ identifies *where* we are evaluating the field, while $\phi(\mathbf{r})$ or $\mathbf{F}(\mathbf{r})$ tells us what *value* the field assigns at that point.

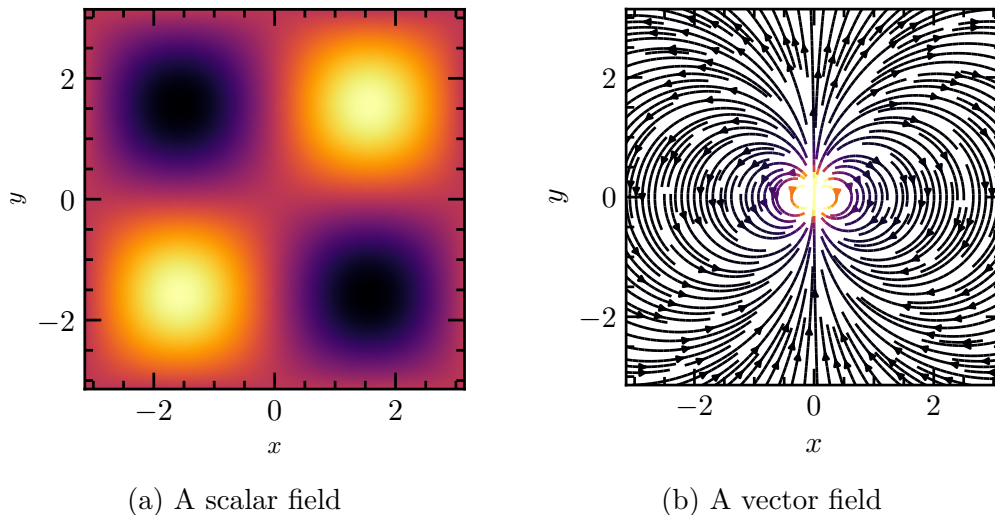


Figure 1: Types of Fields

1.1.5 Differential Calculus

1.1.5.1 The Del Operator

The Del operator ∇ is the generalization of the derivative. We define it as a vector of the partial derivative with respect to the dimension, i.e.

Definition 1.5 : ∇ operator

$$\nabla = \frac{\partial}{\partial x} \hat{\mathbf{i}} + \frac{\partial}{\partial y} \hat{\mathbf{j}} + \frac{\partial}{\partial z} \hat{\mathbf{k}} = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} \quad (1.1.44)$$

As a shorthand, we can represent it as $\frac{\partial}{\partial x_i}$, where x_i is the general dimension. This can be further shorthanded to ∂_i . Thus, a compact vector definition of the Del operator is

$$\nabla = \partial_i \hat{\mathbf{e}}_i \quad (1.1.45)$$

1.1.5.2 Gradient

We often represent fields as $\psi(x, y, z)$; however, we can just make x, y, z into a vector $\mathbf{r} = (x, y, z)$. In that case, the physical interpretation of the gradient becomes clearer.

The **gradient** of a scalar field $\phi(\mathbf{r})$ is a vector field that points in the direction of the greatest rate of increase of ϕ :

Definition 1.6 : Gradient

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{\mathbf{i}} + \frac{\partial\phi}{\partial y}\hat{\mathbf{j}} + \frac{\partial\phi}{\partial z}\hat{\mathbf{k}} = \begin{bmatrix} \frac{\partial\phi}{\partial x} \\ \frac{\partial\phi}{\partial y} \\ \frac{\partial\phi}{\partial z} \end{bmatrix} \quad (1.1.46)$$

It measures how rapidly the scalar field changes and in which direction this change is maximum. For example, in a temperature field $T(x, y, z)$, the gradient ∇T points in the direction where temperature increases most rapidly, and its magnitude gives the rate of increase per unit distance.

In index notation, we can write

$$\nabla\phi = \partial_i\phi\hat{\mathbf{e}}_i, \quad (1.1.47)$$

where ∂_i denotes differentiation with respect to x_i .

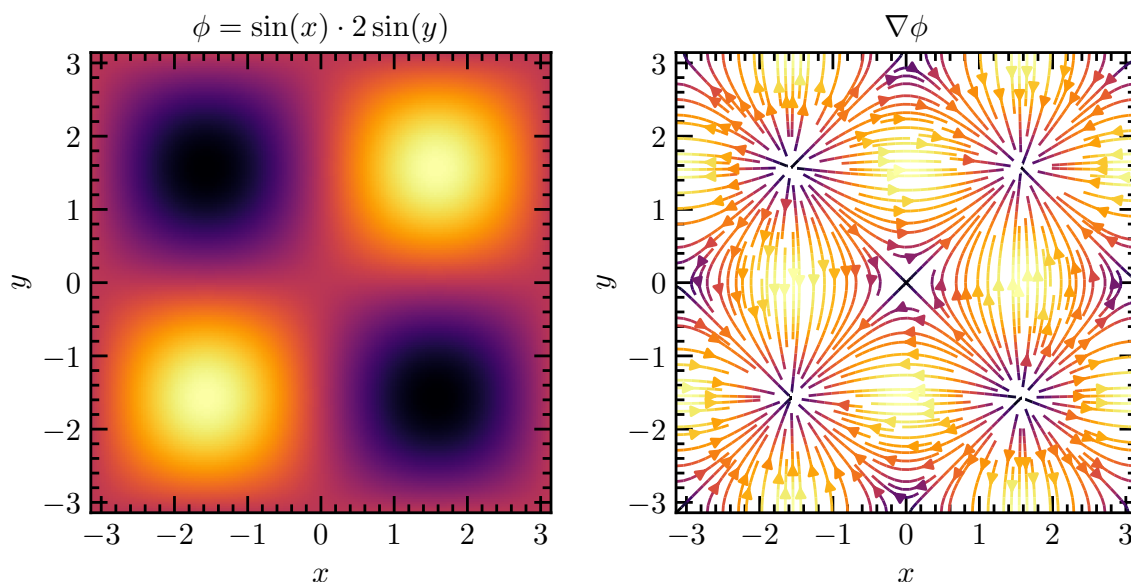


Figure 2: A scalar field and its gradient

1.1.5.3 Divergence

The **divergence** of a vector field $\mathbf{F}(\mathbf{r}) = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = F_i \hat{\mathbf{e}}_i$ measures the net flux per unit volume leaving a point:

Definition 1.7 : Divergence

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \quad (1.1.48)$$

A positive divergence indicates a *source* (more field lines leaving than entering), while a negative divergence indicates a *sink* (field lines converging). If $\nabla \cdot \mathbf{F} = 0$, the field is said to be **solenoidal**, meaning it has no net flux out of any closed surface.

This operation is essentially the dot product of the del operator with the vector field:

$$\nabla \cdot \mathbf{F} = \delta_{ij} \partial_i F_j, \quad (1.1.49)$$

Divergence appears naturally in continuity equations. For instance, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ in Gauss's law for electricity, where it represents the electric charge density acting as a source of the field.

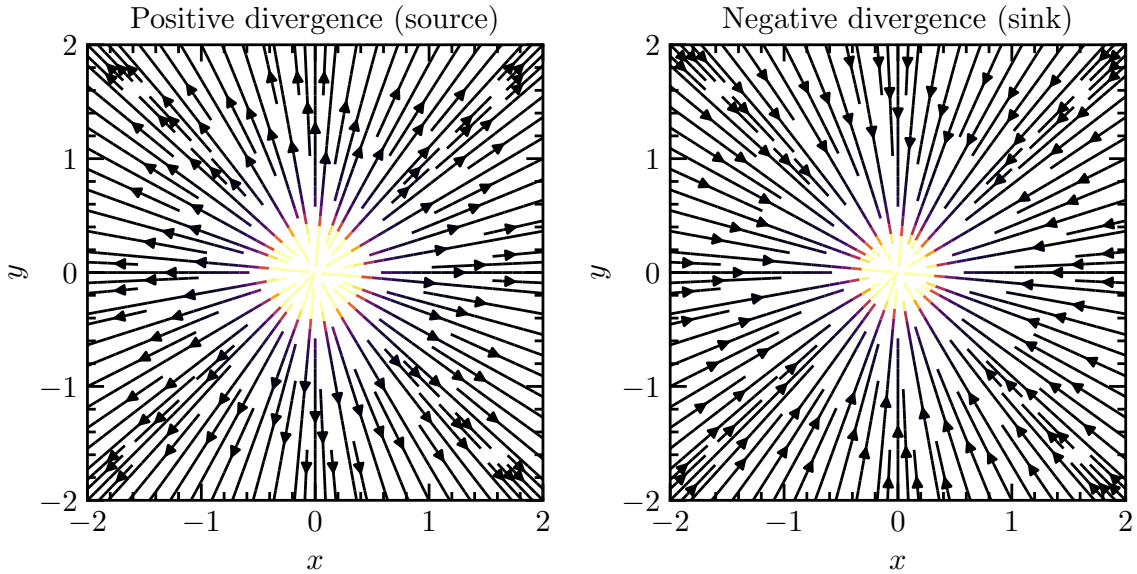


Figure 3: Examples of vector fields with positive and negative divergences

1.1.5.4 Curl

The **curl** of a vector field $\mathbf{F}(\mathbf{r})$ measures its local rotation or the tendency to induce circulation around a point:

Definition 1.8 : Curl

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}. \quad (1.1.50)$$

The direction of $\nabla \times \mathbf{F}$ gives the axis of rotation (using the right-hand rule), and its magnitude gives the rotational strength.

This is equivalent to taking the cross product of the del operator with the vector field:

$$\nabla \times \mathbf{F} = \varepsilon_{ijk} \partial_i F_j \hat{\mathbf{e}}_k \quad (1.1.51)$$

In physical contexts, the curl represents quantities such as the rotational velocity of a fluid or the magnetic field induced by a current. For example, in electromagnetism, $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ expresses Faraday's law of induction.

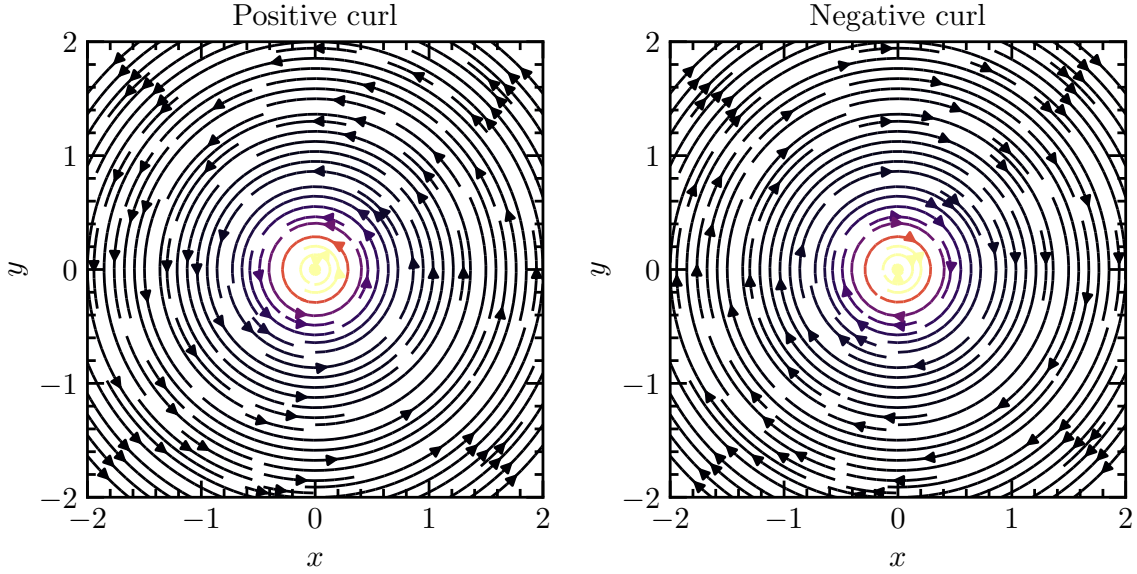


Figure 4: Example of vector fields with positive and negative curl

1.1.5.5 Laplacian

The **Laplacian** operator acts on scalar or vector fields and is defined as the divergence of the gradient:

Definition 1.9 : Laplacian

$$\nabla^2 \phi = \nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}. \quad (1.1.52)$$

It measures how a field value at a point compares to its average in the surrounding region — in other words, how “curved” or “peaked” the field is locally. It is worthy to note that the laplacian of a scalar field is also a scalar field.

In index notation, this becomes

$$\nabla^2 \phi = \delta_{ij} \partial_i \partial_j \phi. \quad (1.1.53)$$

The Laplacian appears in many fundamental physical equations:

- Poisson’s equation: $\nabla^2 \phi = -\rho/\varepsilon_0$ (electrostatics)
- Heat equation: $\partial \phi / \partial t = \alpha \nabla^2 \phi$ (diffusion)
- Schrödinger equation: $\nabla^2 \psi$ governs spatial variations in quantum wavefunctions
- Newtonian Field equation for Gravity: $\nabla^2 \Phi = 4\pi G \rho$

If $\nabla^2 \phi = 0$, the field is called **harmonic**, meaning it satisfies Laplace’s equation and represents an equilibrium distribution such as steady-state temperature or potential.

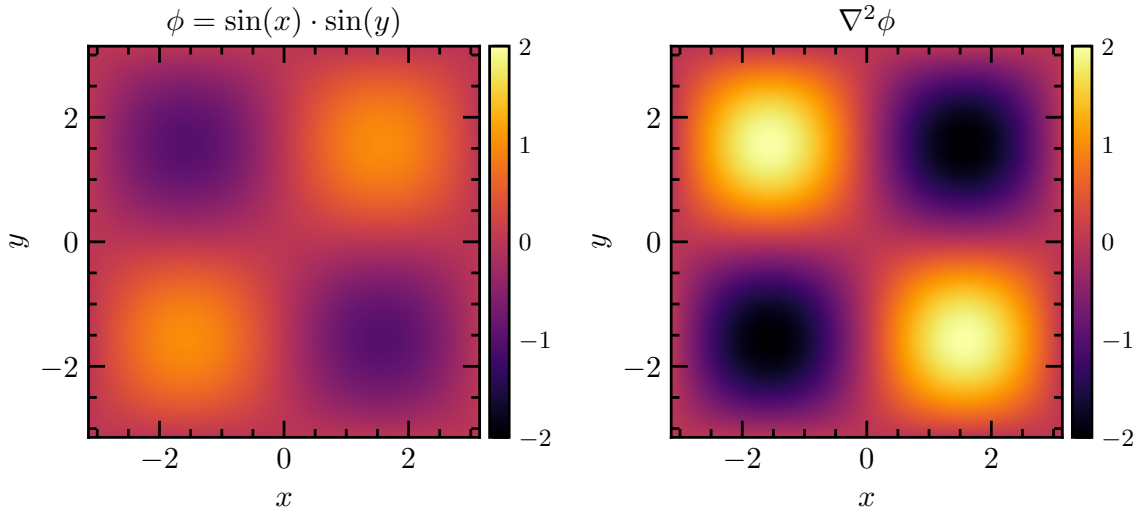


Figure 5: Scalar field and its Laplacian

1.1.5.6 Operations on these Operators

The usual differentiation rules still apply for ∇ , $\nabla \cdot$, and $\nabla \times$, because they are built from ordinary partial derivatives. One only has to account for the fact that the fields may be scalar or vector valued.

Product Rule For scalar fields f, g

$$\nabla(fg) = f \nabla g + g \nabla f \quad (1.1.54)$$

For a scalar f and a vector field $\mathbf{A}$

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + (\nabla f) \cdot \mathbf{A} \quad (1.1.55)$$

and

$$\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) + (\nabla f) \times \mathbf{A} \quad (1.1.56)$$

Chain Rule This also applies for these operations.

Suppose $f = f(u)$ and $u = u(x_1, x_2, x_3)$, then

$$\nabla f(u) = \frac{df}{du} \nabla u \quad (1.1.57)$$

More generally, if $f = f(u, v)$ and $u = u(x, y, z)$ and $v = v(x, y, z)$

$$\nabla f(u, v) = \frac{\partial f}{\partial u} \nabla u + \frac{\partial f}{\partial v} \nabla v \quad (1.1.58)$$

Quotient Rules For scalar fields f, g and vector field $\mathbf{A}$, we have three possible quotient rules.

$$\begin{aligned} \nabla \left(\frac{f}{g} \right) &= \frac{g \nabla f - f \nabla g}{g^2} \\ \nabla \cdot \left(\frac{\mathbf{A}}{g} \right) &= \frac{g(\nabla \cdot \mathbf{A}) - \mathbf{A} \cdot (\nabla g)}{g^2} \\ \nabla \times \left(\frac{\mathbf{A}}{g} \right) &= \frac{g(\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla g)}{g^2} \end{aligned}$$

Additional Rules Some other useful identities are

$$\begin{aligned} \nabla \cdot (\mathbf{A} + \mathbf{B}) &= \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B} \\ \nabla \times (\mathbf{A} + \mathbf{B}) &= \nabla \times \mathbf{A} + \nabla \times \mathbf{B} \end{aligned}$$

And other second derivative identities

$$\begin{aligned} \nabla \cdot (\nabla f) &= \nabla^2 f \\ \nabla \times (\nabla f) &= 0 \\ \nabla \cdot (\nabla \times \mathbf{A}) &= 0 \end{aligned}$$

1.1.6 Integral Calculus

1.1.6.1 Line integrals

For reference, a line integral is defined as the integral over the path Γ of a parameterized vector path function $\mathbf{r}(t)$ over a scalar field ϕ or a vector field $\mathbf{F}$ with respect to a parameter t . Go figure. It comes in two forms: scalar line integrals and vector line integrals.

For a scalar field, the line integral is

Definition 1.10 : Scalar line Integral

$$\int_{\Gamma} \phi(\mathbf{r}) \, ds = \int_a^b \phi(\mathbf{r}(t)) \left\| \frac{d\mathbf{r}}{dt} \right\| dt. \quad (1.1.59)$$

Here, ds is the differential arc length along the curve. Since

$$ds = \left\| \frac{d\mathbf{r}}{dt} \right\| dt, \quad (1.1.60)$$

the scalar line integral can be understood as summing the value of the scalar field along the curve, weighted by the infinitesimal distance traveled along that curve. It is therefore independent of the particular parameterization of the curve, provided that the parameterization preserves its orientation and traces the same path.

For a vector field, the line integral is

Definition 1.11 : Vector line Integral

$$\int_{\Gamma} \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt. \quad (1.1.61)$$

Unlike the scalar line integral, the vector line integral depends on the direction in which the path is traversed. Reversing the orientation of Γ gives

$$\int_{-\Gamma} \mathbf{F} \cdot d\mathbf{r} = - \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r}. \quad (1.1.62)$$

The quantity $d\mathbf{r}$ is the infinitesimal displacement vector along the curve,

$$d\mathbf{r} = \frac{d\mathbf{r}}{dt} dt, \quad (1.1.63)$$

so the vector line integral measures the component of the vector field along the direction of motion. In physics, this is why the work done by a force $\mathbf{F}$ along a path Γ is written as

$$W = \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r}. \quad (1.1.64)$$

If the vector field is conservative, so that

$$\mathbf{F} = -\nabla\phi, \quad (1.1.65)$$

then the integral depends only on the endpoints of the curve rather than on the path taken. In that case, we have the fundamental theorem of gradients.

Theorem 1.1 : Fundamental theorem of Gradients

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{r} = \phi(\mathbf{a}) - \phi(\mathbf{b}), \quad (1.1.66)$$

where $\mathbf{a}$ and $\mathbf{b}$ are the initial and final points of the path. Equivalently, for contour integral of a closed curve (where the endpoints are the same),

$$\oint_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = 0. \quad (1.1.67)$$

This gives a useful connection between line integrals and the gradient: gradients of scalar fields are conservative vector fields. More generally, in a simply connected region,

$$\nabla \times \mathbf{F} = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{F} = \nabla\phi \quad (1.1.68)$$

for some scalar potential ϕ , subject to the appropriate regularity conditions.

1.1.6.2 Surface Integrals

A surface integral is the analogous construction for a two-dimensional surface embedded in three-dimensional space. Let a surface Σ be parameterized by two parameters u and v ,

$$\mathbf{r} = \mathbf{r}(u, v), \quad (u, v) \in D. \quad (1.1.69)$$

The tangent vectors to the surface are

$$\frac{\partial \mathbf{r}}{\partial u}, \quad \frac{\partial \mathbf{r}}{\partial v}, \quad (1.1.70)$$

and therefore the infinitesimal area element is

$$dS = \left\| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right\| du dv. \quad (1.1.71)$$

For a scalar field ϕ , the surface integral is

Definition 1.12 : Scalar surface integral

$$\int_{\Sigma} \phi dS = \iint_D \phi(\mathbf{r}(u, v)) \left\| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right\| du dv. \quad (1.1.72)$$

This is the two-dimensional analogue of the scalar line integral: rather than summing the field along a curve, we sum it over an entire surface, weighted by the infinitesimal area element.

For a vector field $\mathbf{F}$, however, the natural surface integral is a *flux integral*. The oriented surface element is defined by

$$d\mathbf{S} = \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv, \quad (1.1.73)$$

where the direction of $d\mathbf{S}$ is determined by the chosen orientation of the surface. The flux of $\mathbf{F}$ through Σ is then

Definition 1.13 : Vector surface integral

$$\int_{\Sigma} \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv. \quad (1.1.74)$$

The dot product selects the component of the vector field normal to the surface. Thus, the flux measures how much of the field passes through the surface rather than how much of it runs along the surface.

1.1.6.3 Volume Integrals

Finally, a volume integral is an integral over a three-dimensional region V . For a scalar field ϕ , we write

$$\int_V \phi dV. \quad (1.1.75)$$

In Cartesian coordinates,

$$dV = dx dy dz, \quad (1.1.76)$$

so that

Definition 1.14 : Volume integral

$$\int_V \phi dV = \iiint_V \phi(x, y, z) dx dy dz. \quad (1.1.77)$$

These constructions form the basis for the integral versions of the fundamental theorems of vector calculus, including Green's theorem, Stokes' theorem, and the divergence theorem. These theorems relate derivatives of a field within some region to the behavior of that field on the boundary of the region.

1.1.7 Gauss and Stokes Theorem

Gauss's and Stokes' theorems are two of the most important results in vector calculus, providing relationships between surface and volume (or line and surface) integrals. They serve as generalizations of the Fundamental Theorem of Calculus to higher dimensions, allowing one to convert integrals over regions into integrals over their boundaries.

1.1.7.1 Gauss's Divergence Theorem

The divergence theorem, also known as Gauss's theorem, relates the flux of a vector field through a closed surface to the divergence of the field inside the volume enclosed by that surface. Mathematically, it is expressed as

Theorem 1.2 : Gauss' Theorem

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \oiint_S \mathbf{F} \cdot d\mathbf{S}, \quad (1.1.78)$$

where V is the volume enclosed by the closed surface S , $\mathbf{F}$ is a continuously differentiable vector field, and $d\mathbf{S}$ is the outward-pointing area element on S . This theorem is widely used in electromagnetism, fluid mechanics, and gravitational theory.

This is sometimes shortened as

$$\int_{\Omega} \nabla \cdot \mathbf{F} dV = \oint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{S} \quad (1.1.79)$$

1.1.7.2 Stokes' Theorem

Stokes' theorem relates the circulation of a vector field around a closed curve to the curl of the field over the surface bounded by that curve. It can be written as

Theorem 1.3 : Stokes' Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}, \quad (1.1.80)$$

where C is the positively oriented boundary curve of the surface S . Stokes' theorem generalizes the concept of Green's theorem in the plane to three dimensions.

Interpretation In essence, both theorems express how local differential properties of vector fields (divergence and curl) are related to global integral quantities (flux and circulation). They form the mathematical foundation for many physical laws, such as Gauss's law in electromagnetism and the conservation of mass and momentum in fluid dynamics.

1.1.8 Dirac Delta Point Source

The **Dirac delta function** $\delta(\mathbf{r})$ (or $\delta^{(3)}(\mathbf{r})$ for three dimensions) models an idealized point source.

Definition 1.15 : Dirac Delta function

$$\delta(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases} \quad (1.1.81)$$

It has the property

$$\int_{-\infty}^{+\infty} \delta(x) dx = 1 \quad (1.1.82)$$

Its defining property is the sifting integral:

Definition 1.16 : Dirac Delta Sifting

$$\int_V f(\mathbf{r}) \delta(\mathbf{r} - \mathbf{r}_0) d^3r = f(\mathbf{r}_0), \quad (1.1.83)$$

for any test function f and any volume V containing $\mathbf{r}_0$. The delta is zero for $\mathbf{r} \neq \mathbf{r}_0$, singular at $\mathbf{r} = \mathbf{r}_0$, and normalized so that its integral over all space is 1:

$$\int_{\mathbb{R}^3} \delta(\mathbf{r}) d^3r = 1. \quad (1.1.84)$$

Delta and differential operators The Dirac delta often appears as the source term in Poisson-like equations. Two classic identities (in three dimensions) are:

$$\nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta^{(3)}(\mathbf{r}), \quad (1.1.85)$$

$$\nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right) = 4\pi \delta^{(3)}(\mathbf{r}). \quad (1.1.86)$$

These relations are to be understood in the distributional sense: the left-hand side vanishes for $\mathbf{r} \neq \mathbf{0}$ but the integral over any volume containing the origin produces the stated constant.

Example: point charge in electrostatics For an electrostatic potential of a point charge q at the origin,

$$\phi(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \frac{1}{r},$$

Poisson's equation $\nabla^2 \phi = -\rho/\epsilon_0$ is satisfied with $\rho(\mathbf{r}) = q \delta^{(3)}(\mathbf{r})$ because of (1.1.85).

1.1.9 Potential fields

1.1.9.1 Irrotational fields

If the curl of a vector field $\mathbf{F}$ vanishes everywhere, then $\mathbf{F}$ can be written as the gradient of a scalar potential field V .

$$\nabla \mathbf{F} = 0 \iff \mathbf{F} = -\nabla V \quad (1.1.87)$$

Theorem 1.4 : Curl-less or irrotational fields

The following conditions are equivalent.

- (a) $\nabla \times \mathbf{F} = 0$
- (b) $\int_a^b \mathbf{F} \cdot d\mathbf{l}$ is independent of path for any given end points.
- (c) $\oint \mathbf{F} \cdot d\mathbf{l} = 0$
- (d) $\mathbf{F} = -\nabla V$

1.1.9.2 Solenoidal Fields

If the divergence of a vector field $\mathbf{F}$ vanishes everywhere, then $\mathbf{F}$ can be written as the curl of a vector potential field $\mathbf{A}$.

$$\nabla \cdot \mathbf{F} = 0 \iff \mathbf{F} = \nabla \times \mathbf{A} \quad (1.1.88)$$

Theorem 1.5 : Divergence-less or solenoidal fields

The following conditions are equivalent.

- (a) $\nabla \cdot \mathbf{F} = 0$
- (b) $\int \mathbf{F} \cdot d\mathbf{a}$ is independent of surface for any given boundary line.
- (c) $\oint \mathbf{F} \cdot d\mathbf{a} = 0$
- (d) $\mathbf{F} = \nabla \times \mathbf{A}$

Remark :

A vector field potential is not restricted to these definitions. A vector field $\mathbf{F}$ can be written as the sum of a gradient of a scalar and a curl of a vector field.

$$\mathbf{F} = -\nabla V + \nabla \times \mathbf{A} \quad (1.1.89)$$

1.1.10 Curvilinear coordinates

Cartesian coordinates are not the only way to describe position in space. More generally, we may specify a point using three coordinates (q_1, q_2, q_3) whose coordinate surfaces need not be planes. Such a system is called a *curvilinear coordinate system*. The familiar cylindrical and spherical coordinates are important examples.

We introduce an orthogonal curvilinear coordinate system (q_1, q_2, q_3) with mutually perpendicular unit vectors $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$ and scale factors (metric coefficients)

$$h_i(q_1, q_2, q_3) = \left| \frac{\partial \mathbf{r}}{\partial q_i} \right|, \quad i = 1, 2, 3.$$

The scale factor h_i measures the physical distance corresponding to a small change dq_i in the coordinate q_i . Consequently, the infinitesimal displacement is

$$d\mathbf{r} = h_1 dq_1 \hat{\mathbf{e}}_1 + h_2 dq_2 \hat{\mathbf{e}}_2 + h_3 dq_3 \hat{\mathbf{e}}_3$$

The important difference from Cartesian coordinates is that the basis vectors $\hat{\mathbf{e}}_i$ and the scale factors h_i can depend on position. In Cartesian coordinates,

$$h_x = h_y = h_z = 1,$$

and the basis vectors are constant throughout space. In curvilinear coordinates, however, the direction of the basis vectors can change as we move from one point to another.

For an orthogonal coordinate system, a small coordinate volume dq_1, dq_2, dq_3 corresponds to the physical volume

$$dV = h_1 h_2 h_3 dq_1 dq_2 dq_3.$$

Similarly, the area element of a surface on which q_i is constant is

$$dA_i = \frac{h_1 h_2 h_3}{h_i} dq_j dq_k, \quad i \neq j \neq k.$$

Thus, the scale factors determine not only distances but also the geometry of infinitesimal areas and volumes.

Gradient. The gradient of a scalar field ϕ is

Definition 1.17 : Gradient in General Curvilinear Coordinates

$$\nabla \phi = \sum_i \frac{1}{h_i} \frac{\partial \phi}{\partial q_i} \hat{\mathbf{e}}_i \quad (1.1.90)$$

The factors $1/h_i$ are required because $\partial \phi / \partial q_i$ describes the change in ϕ per unit change in the coordinate q_i , whereas the gradient describes change per unit *physical distance*. Since

$$ds_i = h_i dq_i,$$

we have

$$\frac{\partial \phi}{\partial s_i} = \frac{1}{h_i} \frac{\partial \phi}{\partial q_i}.$$

Divergence. The divergence measures the net outward flux of a vector field per unit volume. In orthogonal curvilinear coordinates,

Definition 1.18 : Divergence in general curvilinear coordinates

$$\nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^3 \frac{\partial}{\partial q_i} \left(\frac{h_1 h_2 h_3}{h_i} F_i \right). \quad (1.1.91)$$

The factors involving h_i account for the fact that the areas and volumes associated with the coordinate directions may change from point to point. In Cartesian coordinates, where all $h_i = 1$, this reduces immediately to

Curl. The curl measures the local tendency of a vector field to circulate around a point. For an orthogonal curvilinear coordinate system,

Definition 1.19 : Curl in general curvilinear coordinates

$$\nabla \times \mathbf{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{\mathbf{e}}_1 & h_2 \hat{\mathbf{e}}_2 & h_3 \hat{\mathbf{e}}_3 \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix} \quad (1.1.92)$$

Equivalently, its components are

$$(\nabla \times \mathbf{F})_1 = \frac{1}{h_2 h_3} \left[\frac{\partial}{\partial q_2} (h_3 F_3) - \frac{\partial}{\partial q_3} (h_2 F_2) \right] \quad (1.1.93)$$

$$(\nabla \times \mathbf{F})_2 = \frac{1}{h_3 h_1} \left[\frac{\partial}{\partial q_3} (h_1 F_1) - \frac{\partial}{\partial q_1} (h_3 F_3) \right] \quad (1.1.94)$$

$$(\nabla \times \mathbf{F})_3 = \frac{1}{h_1 h_2} \left[\frac{\partial}{\partial q_1} (h_2 F_2) - \frac{\partial}{\partial q_2} (h_1 F_1) \right]. \quad (1.1.95)$$

Laplacian. The Laplacian of a scalar field is defined as the divergence of its gradient:

$$\nabla^2 \phi = \nabla \cdot (\nabla \phi).$$

Using the gradient and divergence formulas above gives

Definition 1.20 : Laplacian in general curvilinear coordinates

$$\nabla^2 \phi = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^3 \frac{\partial}{\partial q_i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial \phi}{\partial q_i} \right). \quad (1.1.96)$$

This is the standard form of the scalar Laplacian in an orthogonal curvilinear coordinate system.

1.1.10.1 Cylindrical coordinates

Cylindrical coordinates (r, θ, z) describe a point by its distance r from the z -axis, its azimuthal angle θ about the z -axis, and its height z . They are related to Cartesian coordinates by

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$

or equivalently

$$\mathbf{r} = (r \cos \theta, r \sin \theta, z).$$

The corresponding local unit vectors are

$$\begin{aligned}\hat{\mathbf{e}}_r &= (\cos \theta, \sin \theta, 0) \\ \hat{\mathbf{e}}_\theta &= (-\sin \theta, \cos \theta, 0) \\ \hat{\mathbf{e}}_z &= (0, 0, 1)\end{aligned}$$

Unlike Cartesian basis vectors, $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$ depend on θ and therefore change direction from point to point.

The scale factors are

$$h_r = 1 \qquad h_\theta = r \qquad h_z = 1$$

Thus

$$d\mathbf{r} = r dr \hat{\mathbf{e}}_r + r d\theta \hat{\mathbf{e}}_\theta + dz \hat{\mathbf{e}}_z$$

and

$$dV = r dr d\theta dz$$

A vector field is written as

$$\mathbf{F} = F_r \hat{\mathbf{e}}_r + F_\theta \hat{\mathbf{e}}_\theta + F_z \hat{\mathbf{e}}_z$$

Using (1.1.91), its divergence is

$$\nabla \cdot \mathbf{F} = \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} \frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z} \quad (1.1.97)$$

Using (1.1.92), its curl is

$$\nabla \times \mathbf{F} = \left(\frac{1}{r} \frac{\partial F_z}{\partial \theta} - \frac{\partial F_\theta}{\partial z} \right) \hat{\mathbf{e}}_r + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r} \right) \hat{\mathbf{e}}_\theta + \frac{1}{r} \left(\frac{\partial}{\partial r} (r F_\theta) - \frac{\partial F_r}{\partial \theta} \right) \hat{\mathbf{e}}_z \quad (1.1.98)$$

For the scalar Laplacian, $h_1 h_2 h_3 = r$, so

$$\begin{aligned}\nabla^2 \phi &= \frac{1}{r} \left[\frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial \phi}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(r \frac{\partial \phi}{\partial z} \right) \right] \\ &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2}\end{aligned}$$

1.1.10.2 Spherical coordinates

Spherical coordinates (r, θ, φ) describe a point by its radial distance r from the origin, its polar angle θ measured from the positive z -axis, and its azimuthal angle φ measured around the z -axis.

They are related to Cartesian coordinates by

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta,$$

so that

$$\mathbf{r} = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta).$$

The corresponding unit vectors are

$$\begin{aligned} \hat{\mathbf{e}}_r &= (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) \\ \hat{\mathbf{e}}_\theta &= (\cos \theta \cos \varphi, \cos \theta \sin \varphi, -\sin \theta) \\ \hat{\mathbf{e}}_\varphi &= (-\sin \varphi, \cos \varphi, 0) \end{aligned}$$

Again, the basis is position-dependent. In particular, $\hat{\mathbf{e}}_r$ changes whenever either θ or φ changes, while $\hat{\mathbf{e}}_\varphi$ changes as φ changes.

The scale factors are

$$h_r = 1, \quad h_\theta = r, \quad h_\varphi = r \sin \theta.$$

Thus

$$d\mathbf{r} = dr \mathbf{e}_r + r d\theta \hat{\mathbf{e}}_\theta + r \sin \theta d\varphi \hat{\mathbf{e}}_\varphi$$

and

$$dV = r^2 \sin \theta, dr, d\theta, d\varphi$$

A vector field is written as

$$\mathbf{F} = F_r \hat{\mathbf{e}}_r + F_\theta \hat{\mathbf{e}}_\theta + F_\varphi \hat{\mathbf{e}}_\varphi$$

Using (1.1.91), its divergence becomes

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial F_\varphi}{\partial \varphi} \quad (1.1.99)$$

Its curl is

$$\nabla \times \mathbf{F} = \begin{bmatrix} \frac{1}{r \sin \theta} \left(\frac{\partial}{\partial \theta} (\sin \theta F_\varphi) - \frac{\partial F_\theta}{\partial \varphi} \right) \\ \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial F_r}{\partial \varphi} - \frac{\partial}{\partial r} (r F_\varphi) \right) \\ \frac{1}{r} \left(\frac{\partial}{\partial r} (r F_\theta) - \frac{\partial F_r}{\partial \theta} \right) \end{bmatrix} \quad (1.1.100)$$

For the scalar Laplacian, $h_1 h_2 h_3 = r^2 \sin \theta$. Substituting into (1.1.96) gives

$$\begin{aligned} \nabla^2 \phi &= \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} \left(r^2 \sin \theta \frac{\partial \phi}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi}{\partial \theta} \right) + \frac{\partial}{\partial \varphi} \left(\frac{1}{\sin \theta} \frac{\partial \phi}{\partial \varphi} \right) \right] \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \phi}{\partial \varphi^2} \end{aligned}$$

1.1.11 Divergence and the Dirac Delta

As is required from the course notes goals, we aim to prove that the divergence of a radial inverse square vector field is consistent with the Dirac delta point source/sink.

Consider the radial inverse-square vector field

$$\mathbf{F}(\mathbf{r}) = \alpha \frac{1}{r^2} \hat{\mathbf{r}} = \alpha \frac{\mathbf{r}}{r^3} \quad (1.1.101)$$

where $r = \|\mathbf{r}\|$, $\hat{\mathbf{r}} = \mathbf{r}/r$, and $\alpha \in \mathbb{R}$ is constant.

We want to show that the divergence is zero on this field everywhere except at the origin, where it is represented by a Dirac delta:

$$\nabla \cdot \mathbf{F} = 4\pi\alpha\delta^{(3)}(\mathbf{r}) \quad (1.1.102)$$

For $\alpha > 0$, this is a point source, and for $\alpha < 0$, a point sink.

Proof. We prove the relation with the following arguments:

1. Divergence away from the origin

For a purely radial field

$$\mathbf{F} = F_r(r)\hat{\mathbf{r}} \quad (1.1.103)$$

the divergence in spherical coordinates is

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{d}{dr} (r^2 F_r) \quad (1.1.104)$$

Here, the magnitude of F_r is

$$F_r = \frac{\alpha}{r^2} \quad (1.1.105)$$

so,

$$r^2 F_r = \alpha \quad (1.1.106)$$

and therefore, for $r \neq 0$,

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{d\alpha}{dr} \quad (1.1.107)$$

$$= \frac{1}{r^2} \cdot 0 \quad (1.1.108)$$

$$= 0. \quad (1.1.109)$$

Thus,

$$\nabla \cdot \mathbf{F} = 0 \quad (r \neq 0). \quad (1.1.110)$$

Therefore, any nonzero divergence must be concentrated at $r = 0$.

2. Flux through a sphere about the origin

Suppose a sphere S_R of radius R centered at the origin. Since

$$\mathbf{F} = \frac{\alpha}{R^2} \hat{\mathbf{r}} \quad (1.1.111)$$

on the sphere, and

$$d\mathbf{A} = \hat{\mathbf{r}} R^2 \sin \theta d\theta d\phi, \quad (1.1.112)$$

the flux is

$$\oint_{S_R} \mathbf{F} \cdot d\mathbf{A} = \int_0^{2\pi} \int_0^\pi \frac{\alpha}{R^2} R^2 \sin \theta d\theta d\phi, \quad (1.1.113)$$

Evaluating this, we have

$$\oint_{S_R} \mathbf{F} \cdot d\mathbf{A} = \alpha \cdot (4\pi) \quad (1.1.114)$$

Therefore,

$$\oint_{S_R} \mathbf{F} \cdot d\mathbf{A} = 4\pi\alpha \quad (1.1.115)$$

Note that this is independent of R .

3. Gauss' Law

For an ordinary smooth field, the divergence theorem gives

$$\int_V \nabla \cdot \mathbf{F} d^3r = \oint_{\partial V} \mathbf{F} \cdot d\mathbf{A} \quad (1.1.116)$$

Thus, for any volume containig the origin,

$$\int_V \nabla \cdot \mathbf{F} d^3r = 4\pi\alpha \quad (1.1.117)$$

From argument 1, we already established that the divergence is zero everywhere except the origin. Thus, the entire $4\pi\alpha$ must be concetrated at the single point $\mathbf{r} = 0$.

As such, we desire a function $g(\mathbf{r})$ that satisfies

$$g(\mathbf{r}) = 0, \quad \mathbf{r} \neq 0 \quad (1.1.118)$$

but

$$\int_V g(\mathbf{r}) d^3r = 4\pi\alpha \quad (1.1.119)$$

Whenever V contains the origin.

4. Property of the Dirac Delta

The three dimensional Dirac delta is defined by

$$\int_V \delta^{(3)}(\mathbf{r}) d^3r = \begin{cases} 1, & 0 \in V \\ 0, & 0 \notin V \end{cases} \quad (1.1.120)$$

Therefore, $4\pi\alpha\delta^{(3)}(\mathbf{r})$ has the property

$$\int_V 4\pi\alpha\delta^{(3)}(\mathbf{r}) d^3r = \begin{cases} 4\pi\alpha & 0 \in V \\ 0 & 0 \notin V \end{cases} \quad (1.1.121)$$

Which is precisely what we require of our function $g(\mathbf{r})$

Therefore, in the sense of distributions,

$$\nabla \cdot \mathbf{F} = 4\pi\alpha\delta^{(3)}(\mathbf{r}) \quad (1.1.122)$$

Since $\mathbf{F} = \frac{\alpha}{r^2} \hat{\mathbf{r}}$, then we find that

$$\nabla \cdot \left(\alpha \frac{1}{r^2} \hat{\mathbf{r}} \right) = 4\pi\alpha\delta^{(3)}(\mathbf{r}) \quad (1.1.123)$$

We can cancel the α , since it's simply a constant indicating source/sink dynamics. Doing this, we find that

$$\nabla \cdot \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) = 4\pi\delta^{(3)}(\mathbf{r}) \quad (1.1.124)$$

Which is what we set out to prove. $\square$

Thus, we have

Theorem 1.6 : Divergence of a radial inverse quadratic vector field

The divergence of a radial inverse quadratic vector field is consistent with the Dirac delta point source/sink

$$\nabla \cdot \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) = 4\pi\delta^{(3)}(\mathbf{r}) \quad (1.1.125)$$

A particular consequence of this is that the Laplacian of the potential field $\phi = \frac{1}{r}$ (whose gradient field is $\mathbf{F} = \frac{1}{r^2} \hat{\mathbf{r}}$) is also a Dirac delta.

$$\begin{aligned} \nabla \left(\frac{1}{r} \right) &= -\frac{1}{r^2} \hat{\mathbf{r}} \\ \nabla \cdot \nabla \left(\frac{1}{r} \right) &= -\nabla \cdot \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \\ \nabla^2 \left(\frac{1}{r} \right) &= -4\pi\delta^{(3)}(\mathbf{r}) \end{aligned}$$

Theorem 1.7 : Laplacian of an radial inverse potential field

$$\nabla^2 \left(\frac{1}{r} \right) = -4\pi\delta^{(3)}(\mathbf{r}) \quad (1.1.126)$$

1.2 The Electric Field

1.2.1 Definition

The fundamental problem that electrodynamic hopes to solve is this: We have some collection of electric charges $q_1, q_2, q_3, \dots$, called *source charges*; What force do they exert on another charge q_0 , called *test charge*? The positions of the source charges are given (all as functions of time) and the trajectory of the test particle is *to be calculated*. In general, both the source charges and test charge are in motion.

The solution to this problem is facilitated by the **principle of superposition**, which states that the interaction between any two charges is completely unaffected by the presence of others. That means that to determine the force on q_0 , we must first determine the force $\mathbf{F}_1$ due to q_1 alone, then q_2 , and so on. Finally, we take the vector sum of these individual forces.

$$\sum \mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots$$

Thus, if we can find the force on q_0 due to a *single* source charge q , we are, in principle, done, and the rest is just repetition.

A priori, this seems easy. However, upon further study, we find that the force on q_0 not only depends on the separation distance r between the charges, but also depends on both their velocities and on the acceleration of q . Moreover, it is not those states at this instant that matters, but the state that q had in an earlier time, since information travels at a finite speed c , and we can only grasp what happened after the information reached us.

Therefore, in spite of the fact that the basic question (“What is the force on q_0 due to q ?”) is easy to state, it does not pay to confront it head on; rather, we shall go at it by stages. In the meantime, the theory we develop will allow for the solution of more subtle electromagnetic problems that do not present themselves in quite this simple format. To begin with, we shall consider the special case of **electrostatics** in which all the *source charges are stationary* (though the test charge may be moving).

1.2.2 Coulomb’s Law

The force on a test charge q_0 due to a single point charge q , that is at *rest* a distance r away, is given (based on experiments) by **Coulomb’s Law**, a fundamental principle first established by Charles-Augustin de Coulomb in 1785 using a torsion balance experiment.

It is expressed as the following:

Definition 1.21 : Coulomb’s Law

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q_0 q}{r^2} \hat{\mathbf{r}} \quad (1.2.1)$$

The constant ε_0 is called the **permittivity of free space**. In SI units, it is defined as

Definition 1.22 : Permittivity of free space

$$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2 \quad (1.2.2)$$

In other words, then, the force is proportional to the product of the charges, and inversely proportional to the square of their distances². The vector $\mathbf{r}$ is the separation vector from $\mathbf{x}$, the location of q , and to $\mathbf{x}_0$, the location of the q_0

$$\mathbf{r} = \mathbf{x} - \mathbf{x}_0$$

and r is the magnitude of this separation vector, $r = \|\mathbf{r}\|$. The force points along the line from q to q_0 ; it is repulsive if q_0 and q are the same sign, and attractive if their signs are the opposite.

Coulomb's law and the principle of superposition constitute the physical input for electrostatics—the rest, except for some special properties of matter, is mathematical elaboration of these fundamental rules.

1.2.3 The Electric Field

If we have *several* point charges $q_1, q_2, q_3, \dots, q_n$, at distances $r_1, r_2, r_3, \dots, r_n$ from q_0 , then the total force on q_0 is trivially

$$\begin{aligned} \mathbf{F} &= \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots \\ &= \frac{1}{4\pi\varepsilon_0} \left(\frac{q_0 q_1}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_0 q_2}{r_2^2} \hat{\mathbf{r}}_2 + \frac{q_0 q_3}{r_3^2} \hat{\mathbf{r}}_3 + \dots \right) \\ &= \frac{q_0}{4\pi\varepsilon_0} \left(\frac{q_1}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2}{r_2^2} \hat{\mathbf{r}}_2 + \frac{q_3}{r_3^2} \hat{\mathbf{r}}_3 + \dots \right) \end{aligned}$$

or

$$\mathbf{F} = q_0 \mathbf{E} \quad (1.2.3)$$

where

Definition 1.23 : Discrete definition of the Electric field

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i \quad (1.2.4)$$

$\mathbf{E}$ is called the **electric field** of the source charges. Notice that it is a function of position ($\mathbf{r}$) because the separation vectors $\mathbf{r}$ depend on the location of the field point P . But it makes no reference to the test charge q . The electric field is a vector quantity which varies from point to point and is determined by the configuration of source charges. Physically, $\mathbf{E}(\mathbf{r})$ is the force per unit charge that would be exerted on a test charge if you were to place one on P .

²Newton much?

1.2.4 Continuous Charge Distribution

Our definition of the electric field (1.2.4) assumes that the source of the fields is a collection of discrete point charges q_i (Fig 6a). If instead, the charge is distributed continuously over some region, the sum thus becomes an integral

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{\mathbf{r}} dq \quad (1.2.5)$$

There are multiple ways where in charges can be distributed over a region.

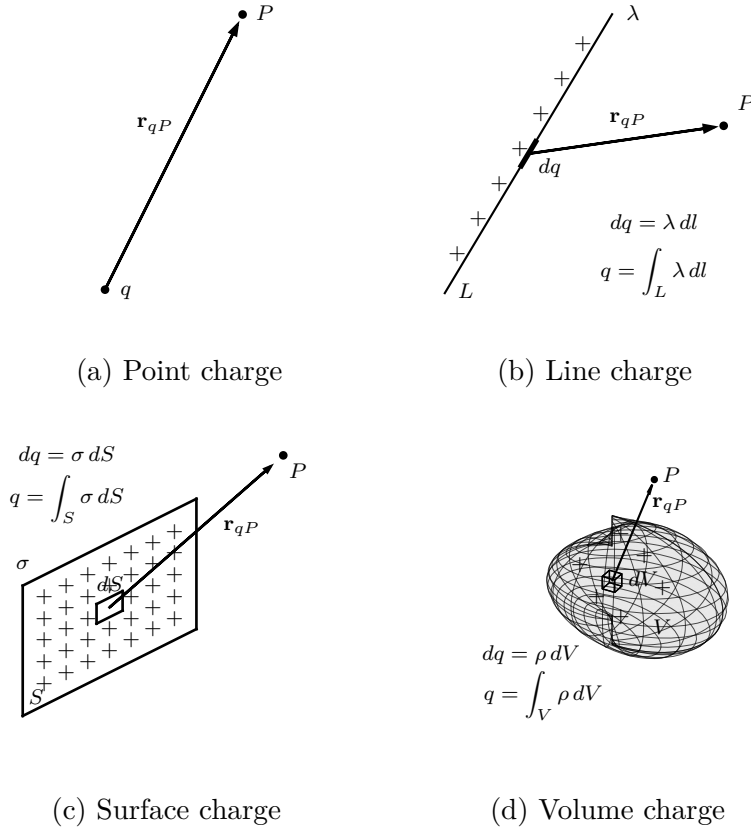


Figure 6: Charge distributions

If the charge is spread out along a *line* (Fig 6b), with the charge-per-unit-length λ , then $dq = \lambda dl$, where dl is a length element along the line. If the charge is smeared over a *surface* (Fig 6c), with charge-per-unit-area σ , then $dq = \sigma dS$, where dS is an area element on the surface. And if the charge fills a *volume* (Fig 6d), with charge-per-unit-volume ρ , then $dq = \rho dV$, where dV is a volume element.

$$dq \rightarrow \lambda dl \sim \sigma dS \sim \rho dV$$

Thus, the electric field of a line charge is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r})}{r^2} \hat{\mathbf{r}} dl; \quad (1.2.6)$$

For a surface charge, we have

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r})}{r^2} \hat{\mathbf{r}} dS; \quad (1.2.7)$$

And for a volume charge

Definition 1.24 : Continuum definition of the Electric Field

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r})}{r^2} \hat{\mathbf{r}} dV; \quad (1.2.8)$$

Equation 1.2.8 is often times referred to as “Coulomb’s Law” because it is such a short step from the original, and because a volume charge is in a sense the most general and realistic case.

1.2.5 Divergence and Curl of Electrostatic Fields

We begin with the simplest possible case of electrostatics. A single point q , situated at the origin.

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

We then have a plot which looks akin to this, the field lines of this point source³

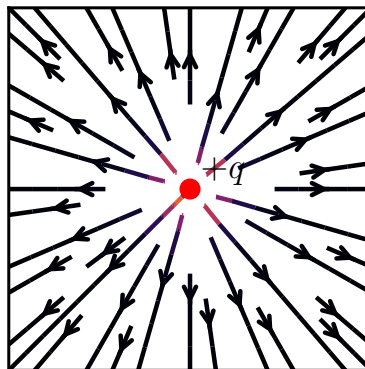


Figure 7: Electric field lines

³Unfortunately `matplotlib` does not have a convenient field line functionality so we have to do `streamplot` for now

1.3 Boundary Conditions, work and energy, conductors

1.4 Calculating Scalar Potentials

1.5 Dielectrics

1.6 Electric Displacement

1.7 Energy in dielectrics, Magnetostatics

1.8 Magnetic Vector Potential

1.9 Unit Summary

Unit 2

Magnetostatics

2.1 Introduction

2.2 Magnetic fields in Matter

2.3 Boundary Value Problems